PRAKTIKUM BASIS DATA

MODUL 03

Scripting SQL Basic Query

Oleh:

Daffa Aufany Febrianto (103122400029)

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL/TP

1. **Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.**

CODE

```
CREATE TABLE mahasiswa (
    id NUMBER PRIMARY KEY,
    nama VARCHAR2(100),
    tempat_lahir VARCHAR2(50),
    tanggal_lahir DATE,
    nomor_hp VARCHAR2(15),
    email VARCHAR2(100),
    tinggi_badan NUMBER,
    berat_badan NUMBER
);
```

OUTPUT

Begin.sql | Welcome Page | Tp2 | MAHASISWA

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexe

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID	NUMBER	No	(null)	1	(null)
2	NAMA	VARCHAR2(100 BYTE)	Yes	(null)	2	(null)
3	TEMPAT_LAHIR	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	TANGGAL_LAHIR	DATE	Yes	(null)	4	(null)
5	NOMOR_HP	VARCHAR2(15 BYTE)	Yes	(null)	5	(null)
6	EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	6	(null)
7	TINGGI_BADAN	NUMBER	Yes	(null)	7	(null)
8	BERAT_BADAN	NUMBER	Yes	(null)	8	(null)

2. **Lakukan insert data pada table nomor 1 sebanyak 15 rows.**

CODE

```
INSERT INTO mahasiswa VALUES (1,'Gatot','Jakarta',TO_DATE('12-05-2003','DD-MM-YYYY'),'081234567890','gatot@gmail.com',170,65);
INSERT INTO mahasiswa VALUES (2,'Zilong','Bandung',TO_DATE('21-08-2002','DD-MM-YYYY'),'081234567891','zilong@gmail.com',168,60);
```

```
INSERT INTO mahasiswa VALUES
(3,'moskov','Surabaya',TO_DATE('10-01-2003','DD-MM-
YYYY'),'081234567892','moskov@gmail.com',160,50);
INSERT INTO mahasiswa VALUES (4,'Jonson','Medan',TO_DATE('04-
03-2002','DD-MM-YYYY'),'081234567893','jonson@gmail.com',172,68);
INSERT INTO mahasiswa VALUES (5,'claude','Padang',TO_DATE('14-
11-2003','DD-MM-YYYY'),'081234567894','claude@gmail.com',158,48);
INSERT INTO mahasiswa VALUES
(6,'Yuzhong','Semarang',TO_DATE('20-02-2002','DD-MM-
YYYY'),'081234567895','yuzhong@gmail.com',175,70);
INSERT INTO mahasiswa VALUES
(7,'LapuLapu','Yogyakarta',TO_DATE('08-06-2003','DD-MM-
YYYY'),'081234567896','lapu@gmail.com',162,52);
INSERT INTO mahasiswa VALUES (8,'Freya','Bekasi',TO_DATE('18-09-
2002','DD-MM-YYYY'),'081234567897','freya@gmail.com',169,64);
INSERT INTO mahasiswa VALUES (9,'Minotour','Bogor',TO_DATE('30-
07-2003','DD-MM-YYYY'),'081234567898','mino@gmail.com',157,49);
INSERT INTO mahasiswa VALUES (10,'Miya','Solo',TO_DATE('12-12-
2002','DD-MM-YYYY'),'081234567899','miya@gmail.com',173,72);
INSERT INTO mahasiswa VALUES (11,'Layla','Depok',TO_DATE('22-
04-2003','DD-MM-YYYY'),'081234567800','layla@gmail.com',159,51);
INSERT INTO mahasiswa VALUES (12,'Harley','Malang',TO_DATE('09-
10-2002','DD-MM-YYYY'),'081234567801','harley@gmail.com',174,69);
INSERT INTO mahasiswa VALUES (13,'Fredrin','Bali',TO_DATE('17-01-
2003','DD-MM-YYYY'),'081234567802','fredrin@gmail.com',161,53);
INSERT INTO mahasiswa VALUES
(14,'Irithel','Makassar',TO_DATE('05-05-2002','DD-MM-
YYYY'),'081234567803','irithel@gmail.com',176,74);
INSERT INTO mahasiswa VALUES (15,'Alice','Palembang',TO_DATE('28-08-
2003','DD-MM-YYYY'),'081234567804','alice@gmail.com',171,67);
```

OUTPUT

Begin.sql | Welcome Page | Tp2 | MAHASISWA

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexes | SQL

Sort.. | Filter:

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Gatot	Jakarta	12-MAY-03	081234567890	gatot@gmail.com	170	65
2	2	Zilong	Bandung	21-AUG-02	081234567891	zilong@gmail.com	168	60
3	3	moskov	Surabaya	10-JAN-03	081234567892	moskov@gmail.com	160	50
4	4	Jonson	Medan	04-MAR-02	081234567893	jonson@gmail.com	172	68
5	5	claude	Padang	14-NOV-03	081234567894	claude@gmail.com	158	48
6	6	Yuzhong	Semarang	20-FEB-02	081234567895	yuzhong@gmail.com	175	70
7	7	LapuLapu	Yogyakarta	08-JUN-03	081234567896	lapu@gmail.com	162	52
8	8	Freya	Bekasi	18-SEP-02	081234567897	freya@gmail.com	169	64
9	9	Minotour	Bogor	30-JUL-03	081234567898	mino@gmail.com	157	49
10	10	Miya	Solo	12-DEC-02	081234567899	miya@gmail.com	173	72
11	11	Layla	Depok	22-APR-03	081234567800	layla@gmail.com	159	51
12	12	Harley	Malang	09-OCT-02	081234567801	harley@gmail.com	174	69
13	13	Fredrin	Bali	17-JAN-03	081234567802	fredrin@gmail.com	161	53
14	14	Irithel	Makassar	05-MAY-02	081234567803	irithel@gmail.com	176	74
15	15	Alice	Palembang	28-AUG-03	081234567804	alice@gmail.com	171	67

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:
a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)

CODE

```
SELECT
nama,
LOWER(nama) AS huruf_kecil,
UPPER(nama) AS huruf_besar,
INITCAP(nama) AS kapital_awal,
CONCAT(nama,email) AS gabung,
LENGTH(nama) AS panjang_nama,
SUBSTR(nama,1,4) AS potong_nama,
LPAD(nama,15,'*') AS lpad_nama,
RPAD(nama,15,'-') AS rpad_nama,
LTRIM('   '||nama) AS ltrim_nama,
RTRIM(nama||'   ') AS rtrim_nama,
INSTR(email,'@') AS posisi_at,
REPLACE(email,'gmail','yahoo') AS ganti_email
FROM mahasiswa;
```

OUTPUT

Script Output × Query Result ×

SQL | All Rows Fetched: 15 in 0.007 seconds

	NAMA	HURUF_KECIL	HURUF_BESAR	KAPITAL_AWAL	GABUNG	PANJANG_NAMA	POTONG_NAMA	LPAD_NAMA	RPAD_NAMA	LTRIM_NAMA	RTRIM_NAMA
1	Gatot	gatot	GATOT	Gatot	Gatotgatot@gmail.com	5	Gato	**********Gatot	Gatot----------	Gatot	Gatot
2	Zilong	zilong	ZILONG	Zilong	Zilongzilong@gmail.com	6	Zilo	*********Zilong	Zilong---------	Zilong	Zilong
3	moskov	moskov	MOSKOV	Moskov	moskovmoskov@gmail.com	6	mosk	*********moskov	moskov---------	moskov	moskov
4	Jonson	jonson	JONSON	Jonson	Jonsonjonson@gmail.com	6	Jons	*********Jonson	Jonson---------	Jonson	Jonson
5	claude	claude	CLAUDE	Claude	claudeclaude@gmail.com	6	clau	*********claude	claude---------	claude	claude
6	Yuzhong	yuzhong	YUZHONG	Yuzhong	Yuzhongyuzhong@gmail.com	7	Yuzh	********Yuzhong	Yuzhong--------	Yuzhong	Yuzhong
7	LapuLapu	lapulapu	LAPULAPU	Lapulapu	LapuLapulapu@gmail.com	8	Lapu	*******LapuLapu	LapuLapu-------	LapuLapu	LapuLapu
8	Freya	freya	FREYA	Freya	Freyafreya@gmail.com	5	Frey	**********Freya	Freya----------	Freya	Freya
9	Minotour	minotour	MINOTOUR	Minotour	Minotourmino@gmail.com	8	Mino	*******Minotour	Minotour-------	Minotour	Minotour
10	Miya	miya	MIYA	Miya	Miyamiya@gmail.com	4	Miya	**********Miya	Miya----------	Miya	Miya
11	Layla	layla	LAYLA	Layla	Laylalayla@gmail.com	5	Layl	**********Layla	Layla----------	Layla	Layla
12	Harley	harley	HARLEY	Harley	Harleyharley@gmail.com	6	Harl	*********Harley	Harley---------	Harley	Harley
13	Fredrin	fredrin	FREDRIN	Fredrin	Fredrinfredrin@gmail.com	7	Fred	********Fredrin	Fredrin--------	Fredrin	Fredrin
14	Irithel	irithel	IRITHEL	Irithel	Irithelirithel@gmail.com	7	Irit	********Irithel	Irithel--------	Irithel	Irithel
15	Alice	alice	ALICE	Alice	Alicealice@gmail.com	5	Alic	**********Alice	Alice----------	Alice	Alice

LTRIM_NAMA	RTRIM_NAMA	POSISI_AT	GANTI_EMAIL
Gatot	Gatot	6	gatot@yahoo.com
Zilong	Zilong	7	zilong@yahoo.com
moskov	moskov	7	moskov@yahoo.com
Jonson	Jonson	7	jonson@yahoo.com
claude	claude	7	claude@yahoo.com
Yuzhong	Yuzhong	8	yuzhong@yahoo.com
LapuLapu	LapuLapu	5	lapu@yahoo.com
Freya	Freya	6	freya@yahoo.com
Minotour	Minotour	5	mino@yahoo.com
Miya	Miya	5	miya@yahoo.com
Layla	Layla	6	layla@yahoo.com
Harley	Harley	7	harley@yahoo.com
Fredrin	Fredrin	8	fredrin@yahoo.com
Irithel	Irithel	8	irithel@yahoo.com
Alice	Alice	6	alice@yahoo.com

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH| YEAR]))

CODE

```
SELECT
nama,
tanggal_lahir,
TO_CHAR(tanggal_lahir,'MM') AS bulan_angka,
TO_CHAR(tanggal_lahir,'MON') AS bulan_singkat,
TO_CHAR(tanggal_lahir,'MONTH') AS bulan_panjang,
TO_CHAR(tanggal_lahir,'DD') AS tanggal,
TO_CHAR(tanggal_lahir,'DY') AS hari_singkat,
TO_CHAR(tanggal_lahir,'YYYY') AS tahun,
TO_CHAR(tanggal_lahir,'YY') AS tahun_pendek,
SYSDATE AS waktu_sekarang,
LAST_DAY(tanggal_lahir) AS akhir_bulan,
ADD_MONTHS(tanggal_lahir,6) AS tambah_6_bulan,
MONTHS_BETWEEN(SYSDATE,tanggal_lahir) AS selisih_bulan
FROM mahasiswa;

SELECT
tanggal_lahir,
NEXT_DAY(tanggal_lahir,'MONDAY') AS senin_berikutnya,
ROUND(tanggal_lahir,'YEAR') AS pembulatan_tahun
FROM mahasiswa;
```

OUTPUT

Script Output | Query Result

SQL | All Rows Fetched: 15 in 0.007 seconds

	NAMA	TANGGAL_LAHIR	BULAN_ANGKA	BULAN_SINGKAT	BULAN_PANJANG	TANGGAL	HARI_SINGKAT	TAHUN	TAHUN_PENDEK
1	Gatot	12-MAY-03	05	MAY	MAY	12	MON	2003	03
2	Zilong	21-AUG-02	08	AUG	AUGUST	21	WED	2002	02
3	moskov	10-JAN-03	01	JAN	JANUARY	10	FRI	2003	03
4	Jonson	04-MAR-02	03	MAR	MARCH	04	MON	2002	02
5	claude	14-NOV-03	11	NOV	NOVEMBER	14	FRI	2003	03
6	Yuzhong	20-FEB-02	02	FEB	FEBRUARY	20	WED	2002	02
7	LapuLapu	08-JUN-03	06	JUN	JUNE	08	SUN	2003	03
8	Freya	18-SEP-02	09	SEP	SEPTEMBER	18	WED	2002	02
9	Minotour	30-JUL-03	07	JUL	JULY	30	WED	2003	03
10	Miya	12-DEC-02	12	DEC	DECEMBER	12	THU	2002	02
11	Layla	22-APR-03	04	APR	APRIL	22	TUE	2003	03
12	Harley	09-OCT-02	10	OCT	OCTOBER	09	WED	2002	02
13	Fredrin	17-JAN-03	01	JAN	JANUARY	17	FRI	2003	03
14	Irithel	05-MAY-02	05	MAY	MAY	05	SUN	2002	02
15	Alice	28-AUG-03	08	AUG	AUGUST	28	THU	2003	03

TAHUN_PENDEK	WAKTU_SEKA...	AKHIR_BULAN	TAMBAH_6_BULAN	SELISIH_BULAN
03	09-MAR-26	31-MAY-03	12-NOV-03	273.91802680704898446833930704898446833 9
02	09-MAR-26	31-AUG-02	21-FEB-03	282.62770422640382317801672640382317801 7
03	09-MAR-26	31-JAN-03	10-JUL-03	277.98254293608124253285543608124253285 5
02	09-MAR-26	31-MAR-02	04-SEP-02	288.17609132317801672640382317801672640 4
03	09-MAR-26	30-NOV-03	14-MAY-04	267.85351067801672640382317801672640382 3
02	09-MAR-26	28-FEB-02	20-AUG-02	288.65996229091995221027479091995221027 5
03	09-MAR-26	30-JUN-03	08-DEC-03	273.04705906511350059737156511350059737 2
02	09-MAR-26	30-SEP-02	18-MAR-03	281.72447841995221027479091995221027479 1
03	09-MAR-26	31-JUL-03	30-JAN-04	271.33738164575866188769414575866188769 4
02	09-MAR-26	31-DEC-02	12-JUN-03	278.91802680704898446833930704898446833 9
03	09-MAR-26	30-APR-03	22-OCT-03	274.59544616188769414575866188769414575 9
02	09-MAR-26	31-OCT-02	09-APR-03	281
03	09-MAR-26	31-JAN-03	17-JUL-03	277.75673648446833930704898446833930704 9
02	09-MAR-26	31-MAY-02	05-NOV-02	286.14383325866188769414575866188769414 6
03	09-MAR-26	31-AUG-03	28-FEB-04	270.40189777479091995221027479091995221

	TANGGAL_LAHIR	SENIN_BERIKUTNYA	PEMBULATAN_TAHUN
1	12-MAY-03	19-MAY-03	01-JAN-03
2	21-AUG-02	26-AUG-02	01-JAN-03
3	10-JAN-03	13-JAN-03	01-JAN-03
4	04-MAR-02	11-MAR-02	01-JAN-02
5	14-NOV-03	17-NOV-03	01-JAN-04
6	20-FEB-02	25-FEB-02	01-JAN-02
7	08-JUN-03	09-JUN-03	01-JAN-03
8	18-SEP-02	23-SEP-02	01-JAN-03
9	30-JUL-03	04-AUG-03	01-JAN-04
10	12-DEC-02	16-DEC-02	01-JAN-03
11	22-APR-03	28-APR-03	01-JAN-03
12	09-OCT-02	14-OCT-02	01-JAN-03
13	17-JAN-03	20-JAN-03	01-JAN-03
14	05-MAY-02	06-MAY-02	01-JAN-02
15	28-AUG-03	01-SEP-03	01-JAN-04

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

CODE

SELECT

```
nama,
TO_CHAR(tanggal_lahir,'DD-MON-YYYY') AS tanggal_format,
TO_CHAR(tinggi_badan) AS tinggi_char,
TO_NUMBER('123') AS contoh_number,
TO_DATE('01-01-2024','DD-MM-YYYY') AS contoh_date
 FROM mahasiswa;
```

OUTPUT

Script Output × | Query Result ×

SQL | All Rows Fetched: 15 in 0.006 seconds

	NAMA	TANGGAL_FORMAT	TINGGI_CHAR	CONTOH_NUMBER	CONTOH_DATE
1	Gatot	12-MAY-2003	170	123	01-JAN-24
2	Zilong	21-AUG-2002	168	123	01-JAN-24
3	moskov	10-JAN-2003	160	123	01-JAN-24
4	Jonson	04-MAR-2002	172	123	01-JAN-24
5	claude	14-NOV-2003	158	123	01-JAN-24
6	Yuzhong	20-FEB-2002	175	123	01-JAN-24
7	LapuLapu	08-JUN-2003	162	123	01-JAN-24
8	Freya	18-SEP-2002	169	123	01-JAN-24
9	Minotour	30-JUL-2003	157	123	01-JAN-24
10	Miya	12-DEC-2002	173	123	01-JAN-24
11	Layla	22-APR-2003	159	123	01-JAN-24
12	Harley	09-OCT-2002	174	123	01-JAN-24
13	Fredrin	17-JAN-2003	161	123	01-JAN-24
14	Irithel	05-MAY-2002	176	123	01-JAN-24
15	Alice	28-AUG-2003	171	123	01-JAN-24

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

CODE

```
SELECT
nama,
tinggi_badan,
berat_badan,
ABS(berat_badan - tinggi_badan) AS selisih,
MOD(tinggi_badan,5) AS sisa_bagi,
FLOOR(berat_badan/3) AS floor_value,
```

```
POWER(tinggi_badan,2) AS pangkat,
ROUND(berat_badan/3,2) AS pembulatan,
CEIL(berat_badan/3) AS ceil_value
 FROM mahasiswa;
```

OUTPUT

Script Output | Query Result

All Rows Fetched: 15 in 0.004 seconds

	NAMA	TINGGI_BADAN	BERAT_BADAN	SELISIH	SISA_BAGI	FLOOR_VALUE	PANGKAT	PEMBULATAN	CEIL_VALUE
1	Gatot	170	65	105	0	21	28900	21.67	22
2	Zilong	168	60	108	3	20	28224	20	20
3	moskov	160	50	110	0	16	25600	16.67	17
4	Jonson	172	68	104	2	22	29584	22.67	23
5	claude	158	48	110	3	16	24964	16	16
6	Yuzhong	175	70	105	0	23	30625	23.33	24
7	LapuLapu	162	52	110	2	17	26244	17.33	18
8	Freya	169	64	105	4	21	28561	21.33	22
9	Minotour	157	49	108	2	16	24649	16.33	17
10	Miya	173	72	101	3	24	29929	24	24
11	Layla	159	51	108	4	17	25281	17	17
12	Harley	174	69	105	4	23	30276	23	23
13	Fredrin	161	53	108	1	17	25921	17.67	18
14	Irithel	176	74	102	1	24	30976	24.67	25
15	Alice	171	67	104	1	22	29241	22.33	23